SL Assignment 1

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Exploratory Data Aanlysis

Table 1: boston data attributes

row_count	column_count
506	13

Table 2: Boston Data Column Attributes

column	DataType	na_count
crim	numeric	0
zn	numeric	0
indus	numeric	0
chas	integer	0
nox	numeric	0
rm	numeric	0
age	numeric	0
dis	numeric	0
rad	integer	0
tax	integer	0
ptratio	numeric	0
lstat	numeric	0
medv	numeric	0

distribution of median house values
0.4
0.2
0.0
−0.2
−0.4
10
20
30
40
50
median house value

- will add more

Model Building

Characteristic	**Beta**	**95% CI**	**p-value**	**q-value**[1]
crim	-0.12	-0.20, -0.04	0.003	0.005
zn	0.04	0.01, 0.07	0.010	0.012
indus	0.02	-0.12, 0.16	0.7	0.8
chas			<0.001	0.001
0	—	—		
1	3.3	1.3, 5.2		
nox	-16	-25, -7.1	<0.001	<0.001
rm	3.9	2.9, 4.8	<0.001	<0.001
age	0.00	-0.03, 0.03	>0.9	>0.9
dis	-1.5	-1.9, -0.98	<0.001	<0.001
rad	0.32	0.17, 0.47	<0.001	<0.001
tax	-0.01	-0.02, -0.01	<0.001	0.001
ptratio	-0.87	-1.2, -0.56	<0.001	<0.001
lstat	-0.59	-0.71, -0.48	<0.001	<0.001

[1]False discovery rate correction for multiple testing
Abbreviation: CI = Confidence Interval
R^2 = 0.730; Adjusted R^2 = 0.722; Sigma = 5.00; Statistic = 87.4; p-value = <0.001; df = 12; Log-likelihood = -1,204; AIC = 2,437; BIC = 2,493; Deviance = 9,659; Residual df = 387; No. Obs. = 400